

Ten-Frames: How Many Do You See?

Answer Key

Kindergarten–Grade 1 Counting and Cardinality

Quick Answers

1. 6	4. 3	7. 6	10. —
2. 9	5. 10	8. 5	
3. 8	6. —	9. —	

Curriculum

Level: Kindergarten–Grade 1 Counting and Cardinality
 K.CC.A.1–K.CC.C.7 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 10 tagged checks.

Common wrong answers

1. *If they got 10: counted the empty boxes as counters too, so the frame looked full*
 5. *If they got 5: counted only the top row and stopped there*
 8. *If they got 7: assumed the top row was full and added five instead of three*
-

1. Full top row, one dot below.

Step 1. The top row of a ten-frame holds five boxes and every one of them is filled, so that row is five without counting it dot by dot.

Step 2. Count on into the bottom row: “six”. That is $5 + 1$.

Step 3. The last number said tells how many. 6

Listen for: a child who goes back to one for the bottom row. The whole point of the frame is that the full row can be trusted.

2. Full top row, four dots below.

Step 1. Top row full, so start at five.

Step 2. Count on: “six, seven, eight, nine” — that is $5 + 4$.

Step 3. One box is still empty, which is a useful check: nine is one less than ten. 9

3. Four above four.

Step 1. Neither row is full here, so the five-shortcut does not apply — count the top row.

Step 2. The top row holds four. Count on through the bottom row: “five, six, seven, eight”. That is $4 + 4$.

Step 3. Four and four make eight. 8

Teaching note: this is the doubles fact $4 + 4$. Seeing it as two equal rows is what makes doubles quick to remember.

4. How many boxes are empty?

Step 1. The question asks about the *empty* boxes, so first find how many are filled: the top row is full and two more sit below, which is $5 + 2 = 7$.

Step 2. The whole frame holds ten, so the empty boxes are what is left after seven: $10 - 7$.

Step 3. Three boxes are empty. 3

Listen for: the answer 7. That is the number of dots, not the number of empty boxes — reread the question together.

5. Both rows full.

Step 1. Each full row is five, and both rows are full.

Step 2. Five and five make ten, and a full ten-frame is ten by definition — that is what the frame is for.

Step 3. There are 10 dots.

6. Comparing Frame A and Frame B.

Step 1. A comparison needs two finished counts, so count each frame before deciding anything. Frame A: $5 + 2 = 7$.

Step 2. Frame B: $5 + 4 = 9$.

Step 3. Seven is less than nine, so the true sentence uses the *less than* sign. $<$

Teaching note: both frames have a full top row, so only the bottom rows can decide it — a first taste of comparing by place.

7. How many more dots to fill the frame?

Step 1. Count what is there: two dots on top and two below, which is $2 + 2 = 4$.

Step 2. A full frame is ten, so the missing dots are $10 - 4$. The empty boxes can also simply be counted: three on the top row and three below.

Step 3. Six more dots are needed. 6

8. The top row is not full.

Step 1. Check the top row before using the five-shortcut — here it holds only three dots, so five would be wrong.

Step 2. Count the top row (three), then count on through the bottom row: “four, five”. That is $3 + 2$.

Step 3. The frame shows 5 dots.

Common wrong answer: 7 — the top row was assumed to be full.

9. Same amount, different arrangement.

Step 1. Count each frame separately. Frame A: $4 + 4 = 8$.

Step 2. Frame B: $5 + 3 = 8$.

Step 3. The two amounts are the same, so neither sign is correct — the true sentence uses equals. $=$

Teaching note: eight looks different in the two frames, and it is still eight. That idea — the arrangement can change while the amount does not — is the heart of this page.

10. Which frame needs more dots?

Step 1. This question is about the *empty* boxes, not the dots, so count the empty boxes in each frame.

Step 2. Frame A shows six dots, so it has $10 - 6 = 4$ empty boxes. Frame B shows nine dots, so it has $10 - 9 = 1$ empty box.

Step 3. Four is more than one, so Frame A needs more dots to be full and the true sentence uses the *greater than* sign. $4 > 1$

Listen for: “Frame B, because it has more dots.” More dots means *fewer* empty boxes — worth acting out with counters.